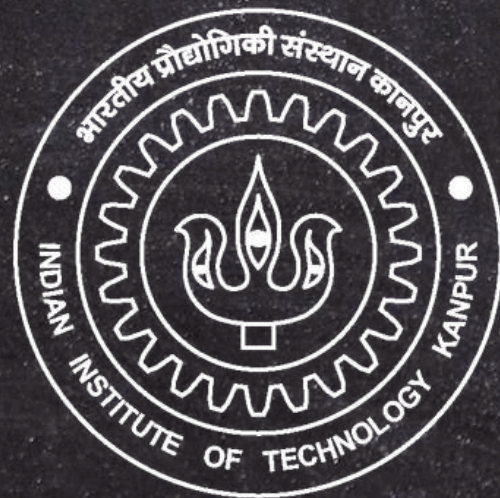




DES643 GROUP PROJECT

IMMERSIVE SPACE EXPLORATION SYSTEM





IMMERSIVE SPACE EXPLORATION SYSTEM

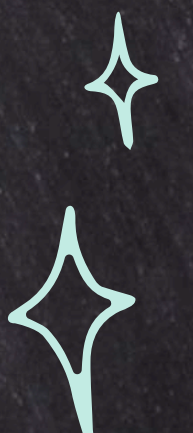
TEAM VIRTUAL

GROUP MEMBERS

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COURSE INSTRUCTOR

PROF. AMAR BEHERA



PROJECT OBJECTIVE

- Build a fully immersive VR system
- Enable users to experience different planets
- Simulate:
 - Atmosphere
 - Scale
 - Physics
 - Interaction

WHAT IS VIRTUAL REALITY (VR)?

- A system that replaces real-world sensory input
- Creates a simulated environment
- Generates a strong sense of **presence**

Presence = feeling of “being there”



WHY VR?

VR enables:

- Embodied learning
- Direct perception of space
- Reduces cognitive load

Improves:

- Understanding
- Engagement

“VR makes 3D relationships directly visible instead of imagined”



DESIGN FOUNDATION

Design Principles Used

- Human-centered design
- Iterative development (Define → Make → Learn)
- Spatial design thinking

Key Idea:

Good VR = Not just graphics → User perception + comfort

SYSTEM OVERVIEW

System Architecture

- Unity-based VR system
- Multiple planetary environments:
 - Mars
 - Venus
 - Saturn
- Components:
 - Terrain
 - Atmosphere
 - Audio
 - UI system



- Binary_Star_Scene
- Mars_terrain
- Mercury_terrain
- Neptune_Scene
- Neutron_Star_Scene
- Saturn_terrain
- Venus_terrain

BINARY STAR SYSTEM

BINARY STAR SYSTEM DESIGN

- The system contains two stars:
 - **Solari** → Yellow star
 - **Lumina** → White star
- These stars are positioned at the two foci of orbital paths
- A total of 5 planets orbit around the combined system



ORBITAL MECHANICS

Orbital Structure

- Planets follow elliptical orbits
- The two stars lie at the foci of the ellipse
- Gravitational influence comes from both stars

System Characteristics:

- Complex orbital paths
- Dynamic spatial relationships
- Non-uniform lighting across planets

VISUAL & EXPERIENCE IMPACT



Dual Lighting System

- Two light sources:
 - Warm yellow (Solari)
 - Cool white (Lumina)

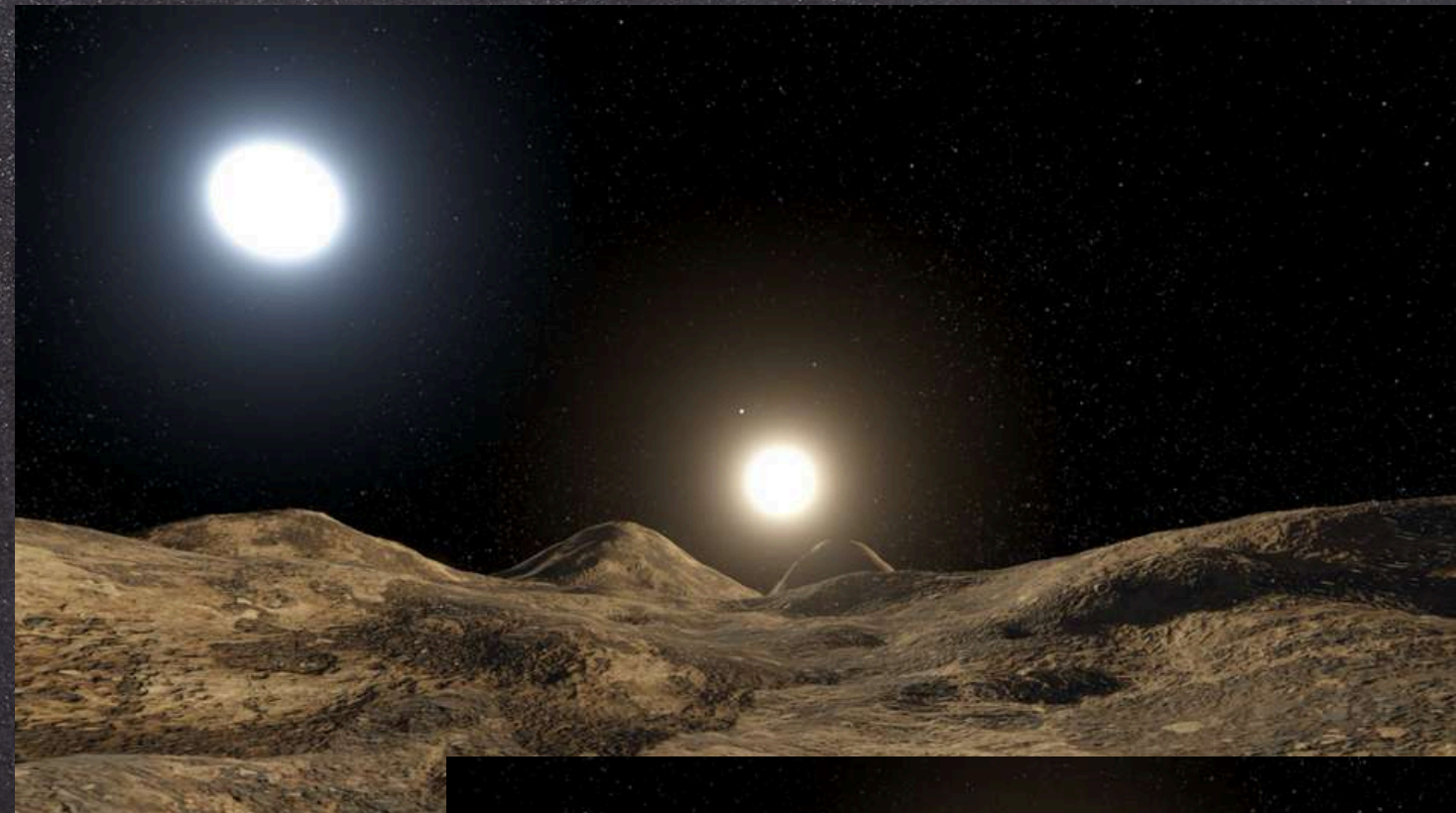
Creates:

- Mixed shadows
- Unique highlights
- Constant variation in lighting

PLANETARY ENVIRONMENTS

PYRONIS

- Closest planet to the binary star system
 - No atmosphere
 - Extreme temperature variation
 - Rocky and cratered
-
- High contrast lighting (very bright)
 - Sharp shadows
 - Barren rocky terrain
 - No fog (clear visibility)



SULVARA

- Thick SO₂ atmosphere
 - High pressure and heat
 - Volcanic surface
 - Yellow/orange sky
-
- Dense fog (low visibility)
 - Volcanic terrain
 - Diffused lighting (no sharp shadows)
 - Heat effects + particles



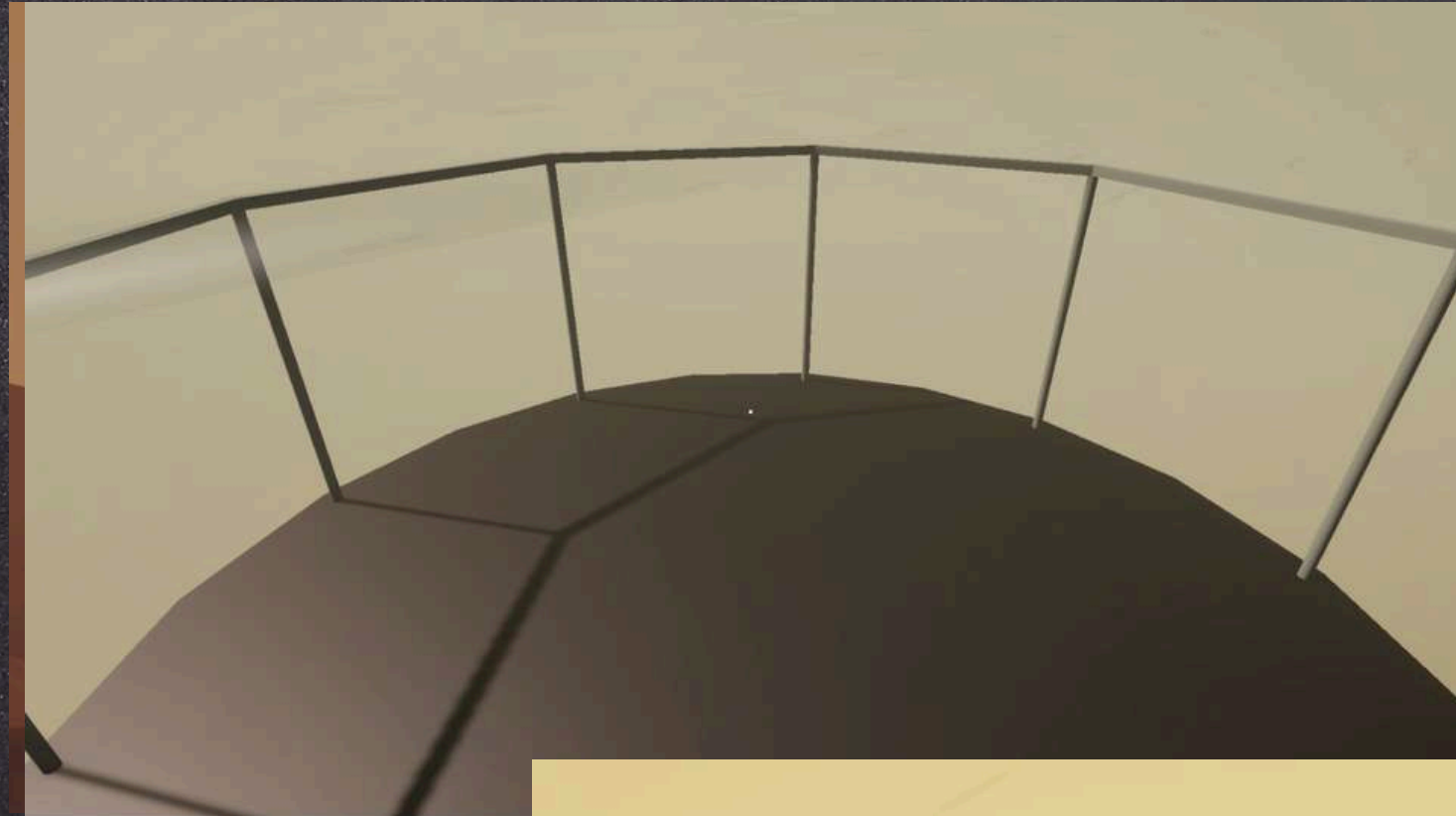
DRAVIX

- Dry surface
- Red dust and iron-rich soil
- Low gravity
- Terrain with craters and hills
- Red sand textures
- Dust particles in air
- Reduced gravity movement



ZEPHYRUS

- Gas giant (no solid surface)
- Thick atmosphere with clouds
- Floating observation platform
- Multi-layer cloud system
- Dense fog for depth
- Continuous wind movement
- Large Dust Particles



THALOR

- Water giant
 - Strong rains and storms
 - Deep blue color
 - Extremely cold
-
- Blue-toned environment
 - Heavy rain sound effects
 - Water bodies like lakes
 - Dynamic atmosphere



AUDIO SYSTEM

Immersive Audio Design

- Ambient sounds (wind, rain)
- Footstep system (dynamic)

Types:

- 2D sound → environment feel
- 3D sound → spatial awareness

PLAYER SYSTEM

Player Interaction

- First-person controller
- Gravity simulation (Mars gravity)
- Movement + grounding logic

Design Principle:

Movement must match perception to avoid sickness

TECH STACK

Tools Used

- Unity Engine
- XR Toolkit
- C# scripting

Game engines handle:

- Physics
- Rendering
- Interaction

KEY CHALLENGES

- Transparency rendering issues
- Depth sorting problems
- VR performance optimization
- Parallax errors

FINAL OUTPUT

- Multi-planet VR experience
- Realistic atmosphere simulation
- Interactive exploration
- Smooth performance

FUTURE SCOPE

- Multiplayer exploration
- AI-driven environments
- NASA data integration
- Procedural planets

CONCLUSION

This project demonstrates:

- How VR can simulate complex planetary environments
- **Using:**
 - Simple geometry
 - Smart rendering
 - Human-centered design

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THANK
YOU!

